

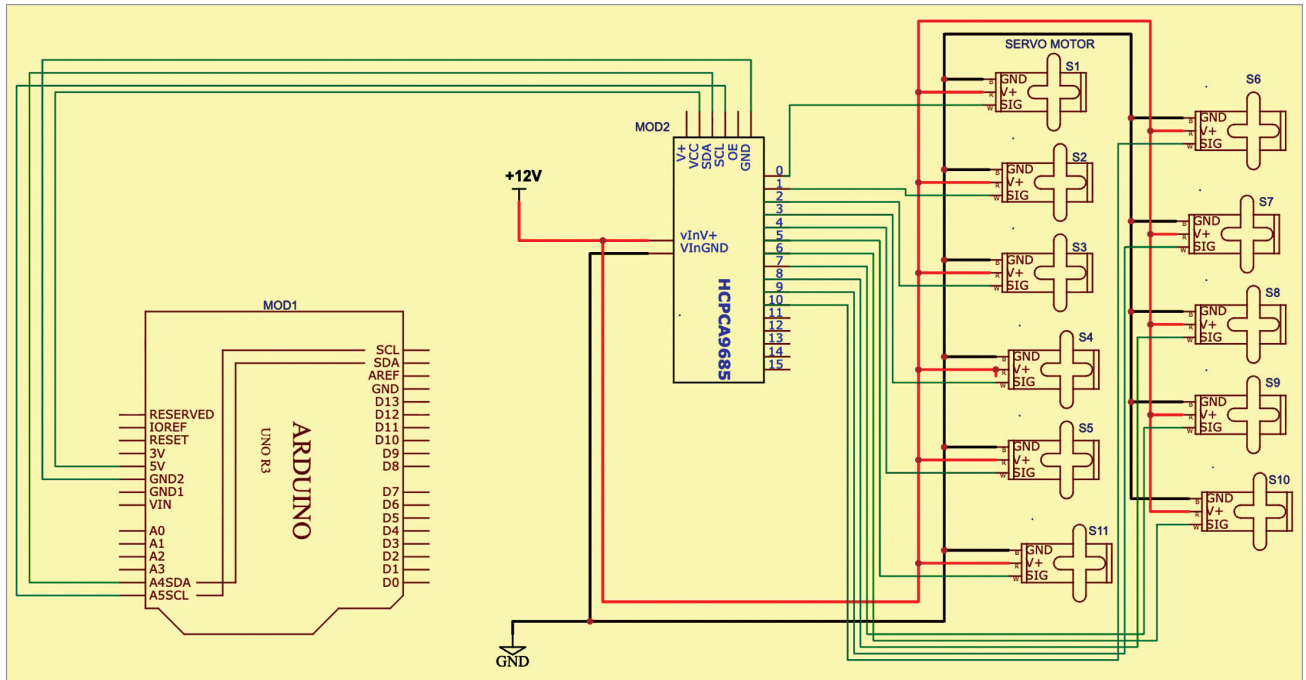


Fig. 2: Circuit diagram of the humanoid robotic arm

Designing and 3D printing

Fig. 4 shows the 3D-printed components of the humanoid robotic arm along with their part names and quantities. All parts are clearly labelled for easy identification and assembly, including finger elements, hand sections, forearm covers, elbow

joints, and shoulder components. Smaller parts such as connectors, axes, and clips enable smooth articulation, particularly for finger movement and gripping. The forearm and arm sections accommodate the servos and internal wiring, while the elbow and shoulder assemblies provide structural support and

rotational motion. All components are designed for straightforward 3D printing and can be assembled using standard fasteners.

Fig. 5 presents the grouped and arranged sections of the 3D-printed robotic arm, including the shoulder, elbow, forearm, hand, and fingers. The design is modular, with each section serving a specific function. The 2-DOF shoulder enables multi-directional movement, the elbow allows bending, and the forearm houses internal components. The hand and fingers enable gripping through coordinated servo action. This modular approach simplifies assembly, testing, and maintenance.

The robot arm parts were first designed in CAD and exported as STL files for 3D printing. These files were then processed using slicer software before printing, as shown in Fig. 4. The parts can be printed using the recommended settings (Table 3) or outsourced to 3D printing service providers.

Assembly of the robot arm

The assembly process is divided into two main parts: first, assembly of the fingers and palm, and second,

```

1  #include <HCPCA9685.h>
2  HCPCA9685 HCPCA9685(0x40); // PCA9685 Controller
3
4  // Definition of pins and limits for each servo
5  const int fingers[5] = {11, 12, 13, 14, 15}; // Finger pins (0° to 180°)
6  const int elbow = 8; // Elbow pin (60° to 200°)
7  const int shoulderFlexion = 3; // Shoulder flexion pin (100° to 180°)
8  const int shoulderLateral = 9; // Shoulder lateral movement pin (130° to 200°)
9
10 // Initial positions (adjustable)
11 const int initialFingersPos = 0; // Fingers open
12 const int initialElbowPos = 60; // Elbow down
13 const int initialShoulderFlexPos = 100; // Shoulder flexed down
14 const int initialShoulderLatPos = 130; // Shoulder lateral centered
15
16 // Speed variables
17 const int speed = 20; // Delay between steps (ms)
18 const int steps = 30; // Steps for smooth movements
19
20 void setup() {
21   HCPCA9685.Init(SERVO_MODE);
22   HCPCA9685.Sleep(false);
23   moveToInitialPosition(); // Set all servos to initial position on startup
24   delay(1000);
25 }
26
27 void moveToInitialPosition() {
28   // Fingers (open hand)
29   for (int i = 0; i < 5; i++) {
30     HCPCA9685.Servo(fingers[i], initialFingersPos);
31   }

```

Fig. 3: Code snippet for setting the servo joint pins and configuring the movement limits for each joint